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ATCTrack: Aligning Target-Context Cues with Dynamic Target States for Robust Vision-Language Tracking

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Abstract

*Vision-language tracking aims to locate the target object in the video sequence using a template patch and a language description provided in the initial frame. To achieve robust tracking, especially in complex long-term scenarios that reflect real-world conditions as recently highlighted by MGIT, it is essential not only to characterize the target features but also to utilize the context features related to the target. However, the visual and textual target-context cues derived from the initial prompts generally align only with the initial target state. Due to their dynamic nature, target states are constantly changing, particularly in complex long-term sequences. It is intractable for these cues to continuously guide Vision-Language Trackers (VLTs). Furthermore, for the text prompts with diverse expressions, our experiments reveal that existing VLTs struggle to discern which words pertain to the target or the context, complicating the utilization of textual cues. In this work, we present a novel tracker named **ATCTrack**, which can obtain multimodal cues **A**ligned with the dynamic target states through comprehensive **T**arget-**C**ontext feature modeling, thereby achieving robust tracking. Specifically, **(1)** for the visual modality, we propose an effective temporal visual target-context modeling approach that provides the tracker with timely visual cues. **(2)** For the textual modality, we achieve precise target words identification solely based on textual content, and design an innovative context words calibration method to adaptively utilize auxiliary context words. **(3)** We conduct extensive experiments on mainstream benchmarks and ATCTrack achieves a new SOTA performance. The code and models will be released at: https://github.com/XiaokunFeng/ATCTrack.*

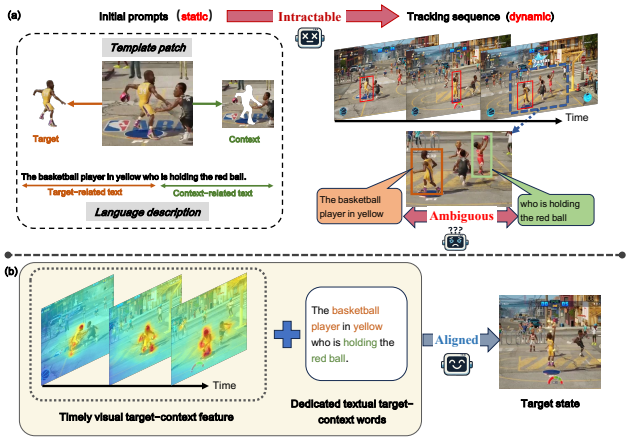


Figure 1. **(a) Limitations of initially given prompts** (*i.e.*, the template patch and language description). Despite containing certain multimodal target-context cues, these static initial prompts are intractable for continuously guiding the tracker in dynamic tracking sequences. Particularly, the objects corresponding to the target-related text and context-related text can be ambiguous, which may mislead the tracker. **(b) Our key insights lie in providing multimodal target-context cues aligned with the dynamic target states.** For the visual modality, we model timely visual target-context features to adapt to dynamic changes. For the textual modality, we achieve precise awareness of target words and mitigate the potential misleading effects of context words.

1. Introduction

Given a template patch and a language description in the initial frame, Vision-Language Tracking (**VLT**) task [49] aims to locate a user-defined object in a video sequence. Harnessing the complementary advantages of multiple modalities [40], recent studies, exemplified by MGIT [33], seek to explore the performance of Vision-Language Trackers (**VLTs**) in **complex long-term** sequences. These scenarios, encompassing multifaceted spatiotemporal and causal

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relationships, more accurately reflect real-world conditions while posing new challenges for tracker design.

To achieve robust tracking within these environments, we first need to utilize the provided visual and textual cues to characterize the **target feature** [48]. As shown in Fig. 1 (a), the appearance of the target in the template image and the target-related text (*e.g.*, target class, appearance attributes) in the language description offer basic reference information for tracking. However, complex long-term scenarios are more commonly accompanied by challenges such as occlusion and similar object distractions [35, 92], making it insufficient to rely solely on these target cues. It is also essential to model the **context feature** of the target [65, 81]. Specifically, the visual content surrounding the target and context-related text (*e.g.*, other reference objects) contribute to more robust tracking. Therefore, the target and context features collectively depict the **target states**, and ***efficiently representing and leveraging target-context features*** is crucial for handling complex long-term scenarios.

Considering that the initially given prompts carry certain multimodal target-context information, most existing VLTs, *e.g.*, VLT_{TT} [30] and JointNLT [98], rely entirely on them for tracking. For the visual template image cropped from the first frame, its cropping region is determined by enlarging the target's bounding box, thus including some context features [93]. For the textual description, the full sentence is utilized as a whole. Although achieving some effectiveness, they overlook the inherent dynamic nature of the target states. In complex long-term scenarios, target states often undergo significant changes. As shown in Fig. 1 (a), the subsequent target states gradually deviate from the initially given prompts. This misalignment makes the latter **intractable** for continuously guiding the tracker. Furthermore, for the text prompts, the object corresponding to the target-related text and context-related text can be **ambiguous**, which may significantly mislead the tracker [33, 44].

Recently, some VLTs, *e.g.*, QueryNLT [65] and TTCTrack [58], attempt to handle different word components of the text prompts specifically, giving sufficient attention to target words and alleviating interference from context words. Although well-motivated, these efforts require trackers to automatically distinguish target and context words through visual-textual feature similarity. Due to the lack of supervision information and the diverse nature of textual expressions, we find that this method does not yield satisfactory results. For an intuitive understanding, Fig. 2 (b) and (c) illustrate cases of correct and incorrect awareness of the target words using this method (see Appendix B.1 for implementation details). Additionally, Fig. 2 (a) presents the quantitative evaluation (see Appendix B.2) results, showing that the classification accuracy for target words achieved by this method is only 29.9%.

To address the above issues, we propose a novel tracker named ***ATCTrack***, which comprehensively models **T**arget-**C**ontext features to obtain multimodal cues **A**ligned with the dynamic target states, thereby achieving robust tracking. Our key insights are illustrated in Fig. 1 (b). For the visual modality, we design an effective temporal visual target-context modeling approach to provide the tracker with timely visual cues. Specifically, we explicitly construct a target-context spatial distribution map and integrate it into the updated temporal memory.

For the textual modality, we first propose a precise target words awareness method based solely on textual content. Intuitively, even without relying on video data, we can identify target words purely from the text. For example, we can determine that the target words in the text prompt of Fig. 1 (a) are "basketball player, yellow." Compared to existing methods that rely on fine-grained alignment between textual words and visual targets [58, 65], our approach simplifies the task and achieves an impressive 96.7% target words classification accuracy (Fig. 2). Given the lack of sentence component labels in existing benchmarks [19, 33, 49, 76], we design an automated target words annotation pipeline by leveraging the off-the-shelf Large Language Models (**LLMs**) [2, 70] to provide supervisory labels for model training. Furthermore, leveraging the accurately identified target words, we design an innovative context words calibration mechanism to alleviate the potential misleading impact of context words cues.

Benefiting from these multimodal target-context cues aligned with dynamic target states, ATCTrack significantly outperforms existing SOTAs on mainstream benchmarks (*i.e.*, MGIT [33], TNL2K [76] and LaSOT_{ext} [20]). Impressively, ATCTrack-B improves over the existing best results by 6.4%, 4.3% and 3.5% in precision, respectively. Our contributions are as follows:

- We propose a novel tracker named ATCTrack, which can obtain multimodal cues aligned with the dynamic target states through comprehensive target-context feature modeling. Compared to the limitations of initial prompts, which typically only align with the initial target state, these aligned cues can guide tracking more robustly.
- For the visual modality, ATCTrack effectively models temporal visual target-context features to capture timely visual cues. For the textual modality, ATCTrack achieves precise awareness of target words and mitigates the potential misleading effects of context words.
- We conduct extensive experiments on mainstream benchmarks and ATCTrack achieves a new SOTA performance.

2. Related Works

2.1. Traditional Vision-Language Tracking

Visual language tracking extends the classic visual single object tracking task by incorporating textual descriptions

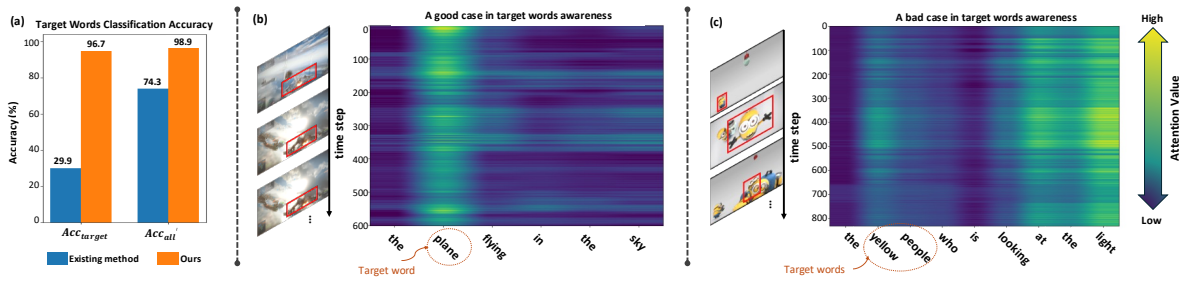


Figure 2. **(a)** Comparison of the existing vision-text similarity-based method and ours in terms of target words classification accuracy. **(b-c)** Attention distribution maps for the target words during tracking using the existing method. In case (b), the model focuses on the word that corresponds to the tracking target, *i.e.*, 'plane,' indicating that the tracker correctly understands the intent embedded in the text prompt, thereby effectively utilizing textual cues. In case (c), the model's focus is on the words 'the light,' which does not match the tracking target 'yellow people,' potentially leading to misguidance in the tracking process. Better viewed in color with zoom-in.

[49]. Many early efforts treat initial prompts as core reference information, utilizing the multimodal target-context cues they contain to guide tracking. Among them, SNLT [23] employs a general language region proposal network to achieve the interaction between multimodal cues and search features, and then uses an aggregation module to integrate multi-scale feature information. To enhance the tracker's ability to align visual and textual modalities, both All-in-One [87] and UVLTrack [56] design specialized contrastive losses [68] to improve the model's multimodal understanding capability. While these VLTs achieve some success, they overly rely on static initial cues and overlook the dynamic nature of the target and its context. This poses a challenge for maintaining robust tracking when the target states deviate from the initial cues [25].

2.2. Object-Context Modeling in Tracking

To accommodate the dynamic changes of target states, many trackers attempt to utilize temporal features [88, 95, 96] to obtain updated target-context cues. For the visual modality, GTI [85] integrates tracking and grounding tasks, replacing the template with updated grounding results. To support longer and denser temporal modeling, some trackers store multiple-step visual temporal features as additional memory. A representative approach involves using RoI features [62] based on predicted bboxes as memory units, *e.g.*, JointNLT [98] and TrDiMP [73]. Given that this localized cropping method can only capture limited visual context information, we represent the global target distribution map and use it to construct temporal visual target-context cues.

In addition to visual modality, recent trackers focus on the unique VLT challenges of misalignment between static textual cues and dynamic target states. MemVLT [25], inspired by prompt learning [1, 42, 97], compresses dynamic target features into a small set of tokens and uses them to implicitly modulate static multimodal cues. To more explicitly leverage multimodal cues, QueryNLT [65] approaches the problem from a target-context perspective, aiming to obtain accurate cues through mutual modulation between dynamic visual features and textual cues. Although QueryNLT shares similar motivations with our work, its key modulation process requires fine-grained alignment between visual targets and textual words in VLTs [89]. As shown in Fig. 2 (a), we conduct a quantitative evaluation and find that this method fails to effectively identify target and context words. In contrast, we design a accurate target words awareness method directly based on textual content and introduce an effective calibration mechanism to mitigate the misleading effects of context words.

3. Methodology

The framework of ATCTrack is depicted in Fig. 3. Given multimodal cues and the search image, the **Text Encoder** and **Vision Encoder** first encode them into specific feature spaces. Then, the **Textual Target-Context Guidance Module** and **Visual Target-Context Guidance Module** sequentially model comprehensive textual and visual target-context features aligned with the dynamic search target, and embed them into the search features. During this process, the **Memory Storage Module** (**MSM**) provides stored visual memory features and saves the updated memory information. Finally, the **Prediction Head** is used to obtain the tracking result based on embedded search features. In the following sections, we will introduce each module in detail.

3.1. Input Encoder

Vision encoder. At time step t, our visual input consists of a search image $x_t \in \mathbb{R}^{3\times H_x\times W_x}$, an initial template patch $z_0 \in \mathbb{R}^{3\times H_z\times W_z}$, and a dynamic template patch $z_t \in \mathbb{R}^{3\times H_z\times W_z}$ updated based on the tracking results [72, 83]. We adopt the one-stream network encoding paradigm [86], which has been widely used by recent mainstream trackers [32, 94, 98]. Specifically, x_t, z_0, and z_t are first projected into token embeddings. Furthermore, inspired by MemVLT [25], we introduce a learnable [CLS] token to capture global visual semantic features [17]. This token is concatenated with template-search tokens and fed into transformer lay-

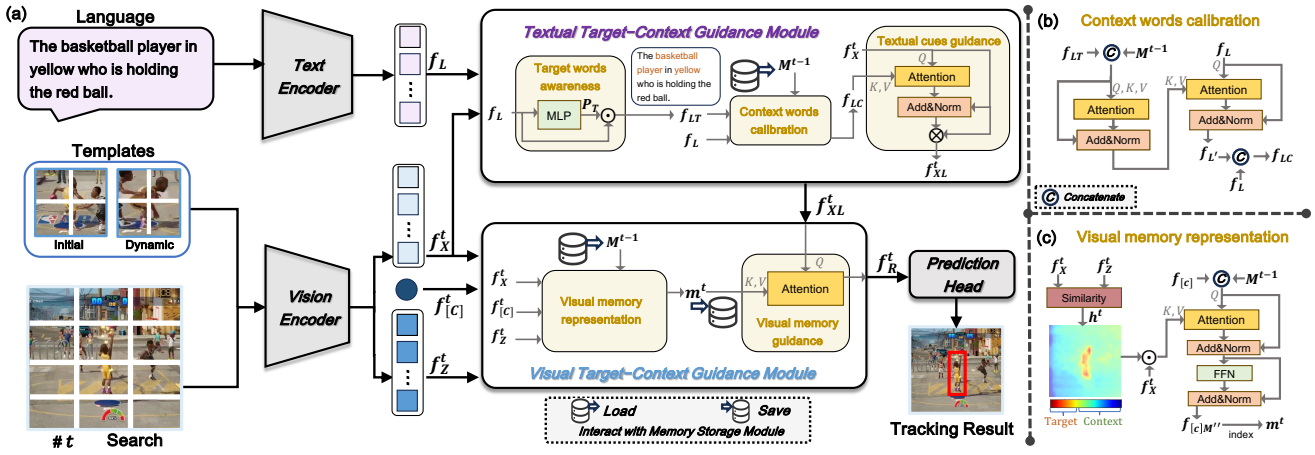


Figure 3. **(a) Framework of our proposed ATCTrack.** Given the language description and template patches as references, ATCTrack locates the target in the search image at time t. The input is first encoded using ***Text*** and ***Vision Encoders***. Subsequently, the ***Textual and Visual Target-Context Guidance Modules*** sequentially embed the aligned textual and visual cues into the search features. During this process, the ***Memory Storage Module*** **(MSM)** provides previously stored memory information (up to $t-1$) and saves updated memory information. Then, the ***Prediction Head*** generates the final tracking results based on the embedded search features. **(b-c) Specific model structures for context words calibration and visual memory representation.**

ers for feature extraction and relational modeling. Finally, we obtain the encoded search feature $f_X^t \in \mathbb{R}^{N_x \times D}$, template feature $f_Z^t \in \mathbb{R}^{N_z \times D}$ corresponding to two template images, and the aggregated [CLS] token $f_{[C]}^t \in \mathbb{R}^{1 \times D}$.

Text encoder. We utilize RoBERTa [17, 52], a classic pre-trained model, as our text encoder. Specifically, for a given sentence, we tokenize it into a sequence of text tokens. The token sequences then are fed into the transformer layers to extract the text embedding feature $f_L \in \mathbb{R}^{N_l \times D}$.

3.2. Textual Target-Context Guidance Module

To fully leverage textual cues, the key insight of this module is to carefully modulate the initial text prompt to adequately focus on the target words and mitigate the potential misleading effects of context words. As shown in Fig. 3, we first explicitly identify the target words from the text content. Then, we design a context words calibration mechanism that leverages the identified target words and the visual memory features provided by the Memory Storage Module (**MSM**) to effectively utilize context words. Finally, we integrate the modulated text features with the search features to achieve the guidance of the textual cues.

Target words awareness. Compared to context words, the information in target words is usually constant and can serve as consistent cues for tracking. However, the diverse nature of textual expressions makes accurately identifying target words challenging. Existing VLTs, such as QueryNLT [65] and TTCTrack [58], attempt to distinguish them based on vision-text similarity. This requirement for fine-grained multimodal alignment increases the learning difficulty. Intuitively, we can identify the target words solely based on the text content. Therefore, we simplify the identification of target words into a multi-label binary classification task, determining whether each word in the sentence belongs to the target words.

Considering the lack of relevant label in existing benchmarks [19, 33, 49, 76], we design specific prompts and leverage the powerful text comprehension capabilities of LLMs [2, 70] to construct an automated and reliable target words annotation pipeline. Due to space constraints, the detailed construction process is described in Appendix A. Leveraging this tailored target words labeling information, we find that accurate target words awareness can be achieved using a lightweight multilayer perceptron built upon the initial text features f_L.

$$p_T = MLP(f_L), \tag{1}$$

where $p_T \in [0,1]^{N_l \times 1}$ denotes the probability of each text token being a target word. By weighting f_L with p_T, we obtain the target words feature $f_{LT} \in \mathbb{R}^{N_l \times D}$.

Context words calibration. The alignment between context words and target states determines whether they guide or mislead the tracker. An intuitive way to determine the timing for utilizing context words is to assess them based on the dynamic evolution of the target states [25, 65]. Therefore, this modeling process requires the temporal memory features stored by the MSM, which represent the latest visual target-context information (details will be described later). Additionally, since context information depends on the target's location, accurately perceiving the target features helps to capture precise context information [4, 9].

Based on these insights, the core of our context words calibration mechanism is to first enhance the target features' representation by integrating identified target words and visual temporal features. These enhanced features are then used to modulate the initial text features to adaptively utilize the context words.

At the current time t, the MSM stores L_m memory units up to time step $t-1$, denoted as $M^{t-1} = \{m_i\}_{i=1}^{L_m}$. As shown in Fig. 3 (b), we first enhance the target feature representation by leveraging the complementarity between the visual memory features M^{t-1} and the textual target words f_{LT}. Specifically, we concatenate the two features:

$$f_{LM} = [f_{LT}; M_*^{t-1}]. \tag{2}$$

Where [;] denotes concatenation along the first dimension, and $M_*^{t-1} \in \mathbb{R}^{L_m \times D}$ is the concatenation of elements in M^{t-1}. Considering the transformer layers' exceptional ability in modeling feature interactions, we apply the vanilla transformer attention operation [71] to f_{LM}:

$$f_{LM'} = Norm(f_{LM} + \Phi_{CA}(f_{LM}, f_{LM})). \tag{3}$$

Here, $\Phi_{CA}(\cdot,\cdot)$ denotes the cross-attention operation, where the first element serves as the query Q and the second element is used to obtain the key K and value V [71]. $Norm$ represents the layer normalization operation.

With the enhancement of target features, the surrounding context information is also indirectly perceived more accurately. Therefore, we use $f_{LM'}$ to modulate the initial static text f_L, allowing for adaptive calibration of the context words. Through the following attention operation, we obtain the meticulously calibrated text features $f_{L'} \in \mathbb{R}^{N_l \times D}$:

$$f_{L'} = Norm(f_L + \Phi_{CA}(f_L, f_{LM'})). \tag{4}$$

Textual cues guidance. Inspired by recent generation models [18, 38, 55] that concatenate various types of text features for utilization, we treat the initial text and the calibrated text as two different types and concatenate them to obtain the comprehensive text feature $f_{LC} = [f_L; f_{L'}]$.

Then, we employ transformer-based cross-attention operations and residual multiplication [94] to obtain the visual search features $f_{XL}^t \in \mathbb{R}^{N_x \times D}$ embedded with textual cues.

3.3. Visual Target-Context Guidance Module

As shown in Fig. 3, this module consists of two core processes: visual memory representation and guidance, aiming to model and leverage dynamic visual target-context memories aligned with the target states. Compared with the sparse dynamic template [82], visual memories can provide denser temporal features. These two mechanisms jointly offer comprehensive temporal information for the tracker. For visual memory representation, we explicitly construct the target-context distribution map and model the memory features at different time steps. Subsequently, we embed the memory cues stored across multiple time steps into the search features, thereby guiding the tracking process.

Visual memory representation. For the [CLS] token $f_{[C]}^t$ encoded by the vision encoder, since it participates in the entire feature integration process of visual information, it serves as a suitable global visual feature representation [17, 25, 88]. Thus, an intuitive approach is to treat $f_{[C]}^t$ as the memory representation at the current time step. However, since the vision encoder is part of the model's early feature modeling stage [86], the precise target location may not have been sufficiently perceived, meaning that $f_{[C]}^t$ lacks explicit awareness of the target-context distribution information. Therefore, we attempt to explicitly construct the target-context distribution heatmap and embed it into $f_{[C]}^t$ to obtain our visual memory.

Specifically, we construct the target-context distribution heatmap by calculating the similarity between the encoded search feature f_X^t and the template features f_Z^t [86]:

$$h^t = (f_X^t \cdot (f_Z^t)^T).mean(dim = 1). \tag{5}$$

Since the f_Z^t are centered on the target, $h^t \in \mathbb{R}^{N_x \times 1}$ reflects the probability of each search token belonging to the target (*i.e.*, not being part of the context). The heatmaps in Fig. 3 (c) are obtained by reshaping h^t into a two-dimensional image space, demonstrating its high interpretability in indicating the spatial distribution of the target-context.

Next, h^t is embedded into $f_{[C]}^t$ to construct the current memory unit m^t. To provide richer target-context dynamic information when constructing m^t, we introduce the historical memory M^{t-1} stored in MSM up to step $t-1$. Note that the L_m memory units in M are obtained through iterative storage of m^t at different time steps. Assuming M^{t-1} has been obtained, we will elaborate on the generation process of each memory unit in the following, while the updating mechanism of M will be presented in Sec. 3.4.

For implementation, we employ the vanilla cross-attention mechanism [71]. In particular, we concatenate the features of $f_{[C]}^t$ and M_*^{t-1} to obtain $f_{[C]M}$ as the query, while using the h^t-weighted f_x^t as both key and value:

$$f_{[C]M'} = Norm(f_{[C]M} + \Phi_{CA}(f_{[C]M}, h^t \odot f_X^t)), \tag{6}$$

$$f_{[C]M''} = Norm(f_{[C]M'} + FFN(f_{[C]M'})). \tag{7}$$

Where $\odot$ represents the Hadamard product, and FFN denotes the feed-forward network. Based on the concatenation index, we extract the feature corresponding to $f_{[C]}^t$ from $f_{[C]M''}$ and utilize it as the memory unit m^t for the current time step. This establishes our memory unit generation mechanism. The generated m^t is then stored in the MSM for subsequent time step computations (see Sec. 3.4).

Method	MGIT (Action)			TNL2K			LaSOT			LaSOT$_{ext}$		
	AUC	P$_{Norm}$	P	AUC	P$_{Norm}$	P	AUC	P$_{Norm}$	P	AUC	P$_{Norm}$	P
Basic Variants												
JointNLT [98]	61.0	78.6	44.5	56.9	73.6	58.1	60.4	69.4	63.6	-	-	-
DecoupleTNL [54]	-	-	-	56.7	-	56.0	71.2	-	75.3	-	-	-
All-in-One [87]	-	-	-	55.3	-	57.2	71.7	82.4	78.5	54.5	63.5	-
MMTrack [94]	-	-	-	58.6	75.2	59.4	70.0	82.3	75.7	49.4	59.9	55.3
QueryNLT [65]	-	-	-	56.9	73.6	58.1	59.9	69.6	63.5	-	-	-
OneTracker [32]	-	-	-	58.0	-	59.1	70.5	79.9	76.5	-	-	-
UVLTrack-B [56]	-	-	-	62.7	-	65.4	69.4	-	74.9	49.2	-	55.8
CTVLT [24]	69.2	-	62.9	62.2	-	79.5	72.3	-	79.7	-	-	-
ChatTracker-B [67]	-	-	-	59.6	76.3	62.1	71.7	80.9	77.5	-	-	-
MemVLT [25]	69.4	81.3	63.7	63.3	80.9	67.4	72.9	85.7	80.5	52.1	63.3	59.8
SUTrack-B224 [14]	-	-	-	65.0	-	67.9	73.2	83.4	80.5	53.1	64.2	60.5
SUTrack-B384 [14]	-	-	-	65.6	-	69.3	74.4	83.9	81.9	52.9	63.6	60.1
ATCTrack-B	73.7	84.5	70.1	67.5	85.3	73.6	74.6	87.0	82.1	54.6	65.7	62.8
Performance-oriented Variants												
ChatTracker-L [67]	-	-	-	65.4	76.5	70.2	74.1	83.8	81.2	-	-	-
UVLTrack-L [56]	-	-	-	64.8	-	68.8	71.3	-	78.3	51.2	-	59.0
SUTrack-L224 [14]	-	-	-	66.7	-	70.3	73.5	83.3	80.9	54.0	65.3	61.7
SUTrack-L384 [14]	-	-	-	67.9	-	72.1	75.2	84.9	83.2	53.6	64.2	60.5
ATCTrack-L	74.0	86.5	76.1	68.6	85.8	75.0	74.7	87.1	82.3	55.4	66.8	64.0

Table 1. Comparison with state-of-the-art trackers on four popular benchmarks: MGIT [33], TNL2K [76], LaSOT [19], and LaSOT$_{ext}$ [20]. The best two results are highlighted in red and blue, respectively.

Visual memory guidance. For the m^t that incorporates both current and historical target-context feature information, we adopt parameter-free attention operations [80] to facilitate its interaction with the search feature f_{XL}^t:

$$f_R^t = softmax(\frac{f_{XL}^t \cdot (m^t)^T}{\sqrt{D}}) \cdot m^t, \tag{8}$$

Here, $f_R^t \in \mathbb{R}^{N_x \times D}$ denotes the search feature that incorporates both textual and visual memory cues, which is subsequently fed into the prediction head.

3.4. Memory Storage Module

As previously mentioned, this module provides the tracker (at t) with stored memory features M^{t-1} (up to $t-1$). Meanwhile, it stores the newly generated memory unit m^t for guiding the tracking process at the next time step $(t+1)$. The L_m memory units in M are initialized with $f_{[C]}^0$ from the first frame and updated using a simple yet widely adopted sliding window approach [7, 80]. For further details, please refer to Appendix C.2.

3.5. Prediction Head and Loss

Based on the search feature f_R^t which integrates multi-modal cues, we utilize a classic CNN-based prediction head [80, 86] to obtain the final bounding box. We employ the focal loss L_{cls} [41], L_1 loss, and the generalized IoU loss L_{iou} [63] to supervise the prediction of bounding box, which are widely used in tracker design[86]. Additionally, for the target words classification task, we employ the binary cross-entropy loss L_{bce} for supervision. The overall loss function is formulated as follows:

$$L_{all} = L_{cls} + 2 \times L_{iou} + 5 \times L_1 + L_2 + 0.2 \times L_{bce}. \tag{9}$$

4. Experiments

4.1. Implementation Details

We adopt RoBERTa-Base [52] as our text encoder and, following recent advanced trackers [14, 66, 80], employ HiViT [69, 90] as our vision encoder. To balance performance and computational efficiency, we develop two model variants: ATCTrack-B and ATCTrack-L, initialized with Fast-iTPN-B and Fast-iTPN-L [69] respectively, with token dimensions D of 512 and 768. The template patches and search images are sized at 128×128 and 256×256, respectively. Additionally, the default length of MSM is set to four, and the dynamic template update strategy follows the STARK [82]. Our tracker is trained on a server with four A5000 GPUs and tested on an RTX-3090 GPU. The tracking speed of ATCTrack-B/L is 35/30 FPS. For a comparison of our model with mainstream VLTs in terms of parameters and speed, please refer to Appendix F.1.

We train our model using the training splits from La-

SOT [19], TNL2K [76], RefCOCOg [57], OTB99-Lang [49], VastTrack [61], GOT-10k [36], and TrackingNet [60]. For GOT-10k [36] and TrackingNet [60] that lack text annotations, we follow the All-in-One's strategy [87] by treating class names as pseudo language labels. Each training sample comprises a text description, two template patches, and four search frames from the same video sequence. Our tracker performs iterative training by sequentially selecting search images. The network parameters are optimized using AdamW optimizer [53] for 150 epochs, with each epoch containing 20,000 randomly sampled instances.

4.2. Comparison with State-of-the-arts

MGIT. MGIT is a latest VLT benchmark specifically tailored for the complex long-term scenarios. Each sequence contains challenging spatio-temporal causal relationships and is annotated with text prompts at three levels of granularity [26, 33, 34, 45–47]. As shown in Tab. 1, ATCTrack demonstrates superior performance at the representative action granularity. Particularly, ATCTrack-B surpassing the SOTA tracker MemVLT [25] by 4.3%, 3.2%, and 6.4% in area under the curve (AUC), normalized precision (P_{Norm}), and precision score (P), respectively. Unlike MemVLT's implicit modulation of static multimodal cues, ATCTrack explicitly adapts these cues from a target-context perspective. These results validate our approach's effectiveness in handling complex long-term scenarios.

TNL2K. TNL2K is also designed for the VLT task, and the introduction of attributes such as "adversarial samples" and "modality switch" significantly adds to the challenges [76]. As shown in Tab. 1, ATCTrack-L outperforms the recent ChatTracker-L by 4.8% in P. Compared to ChatTracker employs multimodal large language models to generate high-quality text annotations for addressing static text limitations, our textual target-context guidance module achieves superior performance with fewer network parameters.

LaSOT & LaSOT$_{ext}$. They are extensions of traditional visual tracking benchmarks [19, 79] by adding text labels, focusing on long-term tracking challenges. Furthermore, LaSOT$_{ext}$ also includes many similar distractors, further complicating the tracking task. As shown in Tab. 1, except for marginally lower AUC and P scores compared to SUTrack-L384, which utilizes larger image resolution and more training datasets, ATCTrack demonstrates highly competitive performance across all other metrics. The outstanding performance across multiple benchmarks further reflects ATCTrack's strong generalization capability.

Qualitative comparison. As shown in Fig. 4, we present the tracking results of ATCTrack-B and two existing SOTA VLTs[25, 56] on four challenging sequences. In these cases, the appearance of the target undergoes significant changes [27, 51], and the language descriptions contain context words that may lead to ambiguity. It is evident that our ATCTrack exhibits greater robustness [11] and effectiveness. More cases can be found in Appendix G.

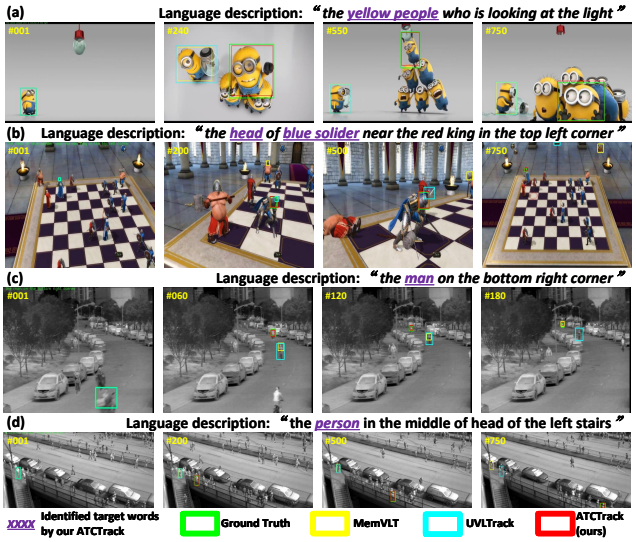


Figure 4. Qualitative comparison results of our tracker with other two VLTs (*i.e.*, MemVLT and UVLTrack) on four challenging cases. Better viewed in color with zoom-in.

#	Setting		TNL2K		LaSOT	
	$Textual_{TC}$	$Visual_{TC}$	AUC	P	AUC	P
1			65.5	70.6	72.5	79.8
2	✓		66.5	72.0	73.6	81.3
3		✓	66.7	72.3	73.7	81.7
4	✓	✓	67.5	73.6	74.7	82.3

Table 2. Ablation study on important model components.

4.3. Ablation Study

To investigate the properties of the various modules in ATCTrack, we conduct comprehensive ablation studies on ATCTrack-B using TNL2K [76] and LaSOT [19]. For the implementation details of each setting, see Appendix E.

Study on important model components. The core contribution of our work is the introduction of a novel multimodal target-context modeling mechanism. In Tab. 2 (#1), we show results using only the initial prompts and the dynamic template to guide tracker. Tab. 2 (#2) and (#3) present results using our textual and visual target-context guidance modules, respectively. Comparatively, Our method shows superior performance with 1.0% and 1.2% AUC gains on TNL2K, respectively, demonstrating the effectiveness of our approach. Additionally, Tab. 2 (#4) indicates that combining these two modules provides complementary benefits, further enhancing tracking performance.

Study on textual target-context modeling. Similar to most VLTs [85, 94, 98], Tab. 3 (#1) (identical to Tab. 2 (#1))

#	Setting	TNL2K		LaSOT	
		AUC	P	AUC	P
1	*naive method*	65.5	70.6	72.5	79.8
2	*+ target words awareness*	65.9	71.7	72.9	80.4
3	*+ context words calibration*	66.5	72.0	73.6	81.3
4	*- dual-type textual guidance*	66.2	71.8	73.2	80.3

Table 3. Ablation study on textual target-context modeling.

#	Setting	TNL2K		LaSOT	
		AUC	P	AUC	P
1	*naive method*	65.5	70.6	72.5	79.8
2	*w RoI*	65.6	71.1	71.1	77.4
3	*w search + crop mask*	65.9	71.4	71.9	78.7
4	*w search + global mask*	66.7	72.3	73.7	81.7

Table 4. Ablation study on visual target-context modeling.

#	Setting	TNL2K		LaSOT	
		AUC	P	AUC	P
1	*ATCTrack-B*	67.5	73.6	74.7	82.3
2	*w/o HiViT backbone*	65.8 (↓ 1.7)	71.3 (↓ 2.3)	72.9 (↓ 1.8)	80.7 (↓ 1.6)
3	*w/o dynamic template*	67.2 (↓ 0.3)	73.0 (↓ 0.6)	73.2 (↓ 1.5)	80.8 (↓ 1.5)
4	*w/o Textual$_{TC}$&Visual$_{TC}$*	65.5 (↓ 2.0)	70.6 (↓ 3.0)	72.5 (↓ 2.2)	79.8 (↓ 2.5)
5	*w/o target words label*	67.0 (↓ 0.5)	72.9 (↓ 0.7)	73.5 (↓ 1.2)	80.6 (↓ 1.7)

Table 5. Ablation study on the contribution of different modules.

treats the text features as a whole, indiscriminately fusing them with visual features. Tab. 3 (#2) and (#3) successively incorporate the target words awareness and context words calibration mechanisms to model and leverage the textual target-context cues aligned with the target state. The progressive improvement in performance demonstrates the effectiveness of our proposed textual guidance approach.

Besides, we utilize both the initial and calibrated textual feature f_{LC} for tracking guidance. Tab. 3 (#4) presents the results of using only the calibrated textual feature $f_{L'}$. Compared to Tab. 3 (#3), the degraded model performance indicates that dual-type textual features help the tracker leverage more comprehensive information.

Moreover, we quantitatively evaluate the target words identification accuracy of our method compared to the existing vision-text similarity-based method (see Appendix B for details). As shown in Fig. 2 (a), despite using only a lightweight multilayer perceptron (Eq. (1)), our method achieves impressive accuracy, distinguishing target words with an overall accuracy (Acc_{all}) of 98.9 %. This facilitates the tracker's utilization of textual target-context cues. Furthermore, our experiments reveal that lightweight text analysis tools, such as Scene Graph Parse [64], demonstrate poor performance in target word recognition, achieving an accuracy ($\text{Acc}_{\text{target}}$) of only 21.0%.

Study on visual target-context modeling. Tab. 4 (#1) shows the results when only sparse initial and dynamic template patches are employed. To model and leverage denser dynamic target-context features, we propose our visual target-context guidance module. Tab. 4 (#3, #4, #5) explores different implementation approaches. Tab. 4 (#3) adopts RoI processing [73], a classic local image feature modeling method, which replaces $h^t \odot f_X^t$ in Eq. (6) with the local region in f_X^t determined by the predicted bbox ($\times 1.5$). Similarly, Tab. 4 (#4) replaces h^t with the local mask of predicted bbox ($\times 1.5$). Compared to Tab. 4 (#2), these two methods achieve slight improvements on TNL2K but suffer significant performance drops on LaSOT. In contrast, our method (*i.e.*, Tab. 4 (#5)) achieves superior results, demonstrating the necessity of representing target-context information from a global perspective.

Further ablation studies. Several components in our model, such as the HiViT backbone [69, 90] and dynamic template [82], are well-established designs for improving tracking performance. Therefore, it is necessary to assess their impact relative to our core contribution, *i.e.*, the multimodal target-context guidance modules. As shown in Tab. 5 (#2, #3), replacing HiViT with ViT [86] or removing the dynamic template leads to performance degradation, which aligns with existing findings[80, 82]. However, Tab. 5 (#4) reveals that removing our multimodal target-context guidance module results in more significant performance drops, indicating that ATCTrack's superior performance primarily stems from our proposed method.

Furthermore, ATCTrack employs unique target word supervision labels, which constitute a significant contribution of our method. To ensure fairness, Tab. 5 (#5) presents the results without this label. Although the performance of the model decreases, it still surpasses the SOTA trackers represented by MemVLT [25] and SUTrack-B224 [14].

5. Conclusion

In pursuit of robust vision-language tracking, especially in complex long-term scenarios that reflect real-world conditions as exemplified by MGIT, we propose a novel tracker named ATCTrack. Through comprehensive target-context modeling, we obtain multimodal cues aligned with dynamic target states, offering an innovative solution to the obstacle where initial static cues fail to provide sustained guidance. For the textual modality, we introduce a precise target words awareness method to ensure sufficient attention to target words, and design an innovative context words calibration mechanism to mitigate potential misleading effects. For the visual modality, we efficiently characterize temporal target-context features to provide timely visual cues for tracking. Through these combined efforts, our model achieves outstanding performance, significantly surpassing existing methods across four mainstream benchmarks.

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